

Code2Create Challenge – Round 3 Problem Statement

Hedge Fund Risk Modeling & Semi-Automated Trading System

Problem Overview:

Design a system that analyzes financial datasets to model risk and simulate a semi-automated trading strategy. The system should generate trading signals and evaluate portfolio performance with proper risk management.

Inputs:

- Market Data: Historical prices, volume, volatility
- Optional: Macroeconomic indicators, sentiment data
- Portfolio Constraints: Initial capital, position limits, risk tolerance

Constraints:

- Semi-automated strategies (rule-based or ML-driven)
- Must include risk modeling techniques
- Simulate realistic conditions (transaction costs, slippage)
- Periodic portfolio rebalancing
- Multi-asset scalability

Objectives:

- Maximize risk-adjusted returns
- Minimize drawdowns and volatility
- Build explainable strategies
- Provide insights via dashboard

Key Metrics:

- Returns (cumulative, annualized)
- Volatility and max drawdown
- Sharpe Ratio
- Beta and Alpha
- Portfolio allocation and trade logs

Evaluation Criteria:

- Performance (Sharpe, alpha)
- Risk management
- Strategy design
- Dashboard quality
- Scalability
- Code quality

Expectations:

Participants should build a full pipeline including preprocessing, feature engineering, signal generation, and dashboard visualization.

Notes:

- No single correct solution
- Focus on risk-aware design
- Simplicity preferred over unnecessary complexity